

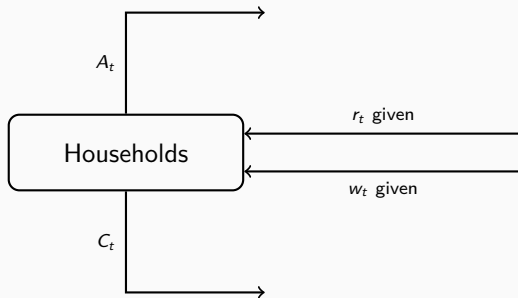
4. Stationary Equilibrium

Adv. Macro: Heterogeneous Agent Models

Raphaël Huleux

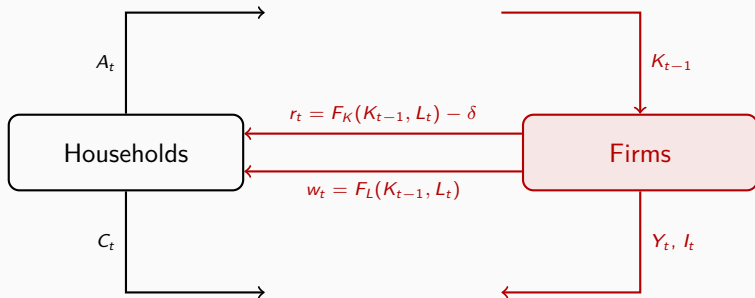
2026

So far: the household block



Partial equilibrium: households take r_t and w_t as given and choose c_t and a_t .

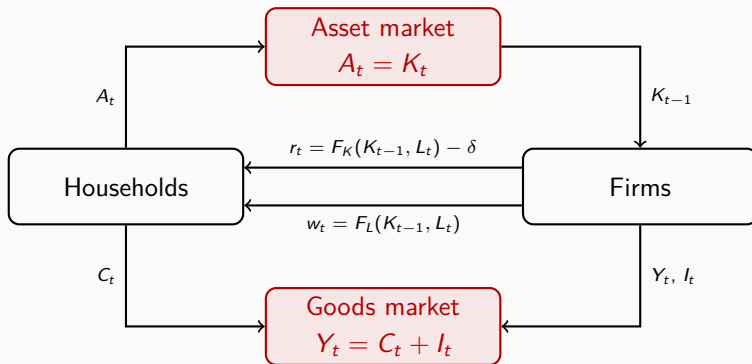
Today: prices come from firms



Firms: r_t and w_t are now determined by the capital stock.

But who decides K_t ?

Today: markets close the model



General equilibrium: capital is what households save, $K_t = A_t$.

Prices must be such that both markets clear: a fixed point.

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Today and next week: use those methods to solve the steady-state of a simple heterogeneous-agent model.

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 2. No interactions

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 1. What determines income and wealth inequality *in the long run*?
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 - **Code:** Based on the **GEModelTools** package
 1. Is in active development
 2. You can help to improve interface, find bugs and features
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- **Literature:** Aiyagari (1994)

Outline of this lecture

1. Recap of the Ramsey (Neo-Classical) model
2. Overview of the Heterogeneous-Agent Neo-Classical model (HANC)
3. How to compute the stationary equilibrium
4. Some economic properties of the HANC stationary equilibrium

Ramsey-recap

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- **Now**: Recap of the Ramsey model

- **Production function:** $Y_t = F(\Gamma_t, K_{t-1}, L_t)$ [capital chosen in $t - 1$ is used for production at t]
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- **Profits:** $\Pi_t = Y_t - w_t L_t - r_t^K K_{t-1}$
- **Profit maximization:** $\max_{K_{t-1}, L_t} \Pi_t$
 1. Rental rate: $\frac{\partial \Pi_t}{\partial K_{t-1}} = 0 \Leftrightarrow r_t^K = F_K(\Gamma_t, K_{t-1}, L_t)$
 2. Real wage: $\frac{\partial \Pi_t}{\partial L_t} = 0 \Leftrightarrow w_t = F_L(\Gamma_t, K_{t-1}, L_t)$

With CRS we get zero profits: $\Pi_t = 0 \Rightarrow$

$$Y_t = w_t L_t + r_t^K K_{t-1} \text{ [functional income distribution]}$$

Ramsey: Zero-profit mutual fund

- Introduce **mutual fund**
 - Takes savings A_{t-1} from households and invest them in available assets
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- **Balance sheet:**

$$A_{t-1} = K_{t-1}$$

- **Utility maximization:**

$$v_0(A_{-1}^{hh}) = \max_{\{C_t^{hh}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(C_t^{hh})$$

s.t.

$$C_t^{hh} + A_t^{hh} = (1 + r_t)A_{t-1}^{hh} + w_t L_t^{hh}$$

Exogenous labor supply: $L_t^{hh} = 1$

- **Euler-equation** (implied by Lagrangian):

$$u'(C_t^{hh}) = \beta(1 + r_{t+1})u'(C_{t+1}^{hh})$$

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- **Walras:** Capital and labor market clears $\Rightarrow$ goods market clears.

Start from

$$\begin{aligned}C_t^{hh} + A_t^{hh} &= (1 + r_t)A_{t-1}^{hh} + w_t L_t^{hh} \\ \Leftrightarrow C_t^{hh} + I_t &= [(1 + r_t)A_{t-1}^{hh} + w_t L_t^{hh} - A_t^{hh}] + (K_t - (1 - \delta)K_{t-1}) \\ &= [(1 + r_t)K_{t-1} + w_t L_t - K_t] + (K_t - (1 - \delta)K_{t-1}) \\ &= r_t^K K_{t-1} + w_t L_t \\ &= Y_t\end{aligned}$$

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- **Note:** Means that we can check if we have solved the numerical model correctly by:
 - Impose two of the market clearing conditions
 - Then check the third market clearing condition (should be zero)

Ramsey: Summary

- Extended form:

$$r_t^K = F_K(\Gamma_t, K_{t-1}, L_t)$$

$$w_t = F_L(\Gamma_t, K_{t-1}, L_t)$$

$$r_t = r_t^K - \delta$$

$$A_t = K_t$$

$$A_t^{hh} = (1 + r_t)A_{t-1}^{hh} + w_t L_t^{hh} - C_t^{hh}$$

$$u'(C_t^{hh}) = \beta(1 + r_{t+1})u'(C_{t+1}^{hh})$$

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- Simplified form:

$$u'(C_t^{hh}) = \beta(1 + F_K(\Gamma_{t+1}, K_t, 1) - \delta)u'(C_{t+1}^{hh})$$

$$K_t = (1 - \delta)K_{t-1} + F(\Gamma_t, K_{t-1}, 1) - C_t^{hh}$$

Ramsey: As an equation system

Eqs. system with unknowns $\{K_t, L_t, r_t^K, w_t, r_t, A_t, A_t^{hh}, C_t^{hh}\}_{t=0}^{\infty}$ and eqs:

$$\begin{bmatrix} r_t^K - F_K(\Gamma_t, K_{t-1}, L_t) \\ w_t - F_L(\Gamma_t, K_{t-1}, L_t) \\ r_t - (r_t^K - \delta) \\ A_t - K_t \\ A_t^{hh} - ((1 + r_t)A_{t-1}^{hh} + w_t L_t^{hh} - C_t^{hh}) \\ u'(C_t^{hh}) - \beta(1 + r_{t+1})u'(C_{t+1}^{hh}) \\ A_t - A_t^{hh} \\ L_t - L_t^{hh} \\ \forall t \in \{0, 1, \dots\}, \text{ given } K_{-1} \end{bmatrix} = 0$$

Ramsey: Steady state

- **Euler-equation** can be solved for r_{ss} and hence K_{ss} :

$$u'(C_{ss}) = \beta(1 + F_K(\Gamma_{ss}, K_{ss}, 1) - \delta)u'(C_{ss}) \Leftrightarrow$$
$$F_K(\Gamma_{ss}, K_{ss}, 1) = \frac{1}{\beta} - 1 + \delta$$

- **Accumulation equation + goods mkt. clearing** then implies C_{ss} :

$$K_{ss} = (1 - \delta)K_{ss} + F(\Gamma_{ss}, K_{ss}, 1) - C_{ss} \Leftrightarrow$$
$$C_{ss} = (1 - \delta)K_{ss} + F(\Gamma_{ss}, K_{ss}, 1) - K_{ss}$$

- Important thing to note: $r_{ss} = 1/\beta - 1$ whatever the level of K_{ss} .
The steady-state asset supply of households is *perfectly elastic*, a horizontal line at $1/\beta - 1$!

HANC



- **Model blocks:**

1. **Firms:** Rent capital from mutual fund and hire labor from the households, produce with given technology, and sell output goods
2. **Zero-profit mutual funds:** Own capital and rent it to firms, take deposits and pay return to household
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3. The Standard Incomplete Market (SIM) model

Heterogeneous households

- **Utility maximization** for household i :

$$v_0(z_{i0}, a_{i,-1}) = \max_{\{c_{it}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{it})$$

s.t.

$$\ell_{it} = z_{it}$$

$$a_{it} = (1 + r_t)a_{it-1} + w_t \ell_{it} - c_{it}$$

$$\log z_{it+1} = \rho_z \log z_{it} + \psi_{it+1}, \quad \psi_{it} \sim \mathcal{N}(\mu_\psi, \sigma_\psi), \quad \mathbb{E}[z_{it}] = 1$$

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- **Incomplete markets due to borrowing constraint**

(fancy words: partial self-insurance, lack of Arrow-Debreu securities)

- **Value function** (at decision)

$$v(z_{it}, a_{it-1}) = \max_{c_{it}} u(c_{it}) + \beta \mathbb{E} [v(z_{it+1}, a_{it})]$$

s.t.

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Distributions and aggregates

- Household policy function x^* where $x \in \{a, c, \ell\}$ function of:
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- Aggregate policy are obtained combining policies and distributions:

$$X_t^{hh}(\{r_\tau, w_\tau\}_{\tau \geq 0}) = \int x_t^*(z_{it}, a_{it-1}, \{r_\tau, w_\tau\}_{\tau \geq t}) d\mathbf{D}_t$$

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- When aggregating we **integrate** out individual states
 - Aggregate X_t^{hh} is only a function of the whole path $\{r_\tau, w_\tau\}_{\tau \geq 0}$ (and of $\mathbf{D}_0$)
- $\Rightarrow$ If we know the path of prices $\{r_t, w_t\}$ we can calculate aggregate household behavior (consumption or savings)

Equation system

$$\begin{bmatrix} r_t^K - F_K(\Gamma_t, K_{t-1}, L_t) \\ w_t - F_L(\Gamma_t, K_{t-1}, L_t) \\ r_t - (r_t^K - \delta) \\ A_t - K_t \\ A_t - A_t^{hh} \\ L_t - L_t^{hh} \\ A_t^{hh} - \int a_t d\mathbf{D}_t \\ L_t^{hh} - \int \ell_t d\mathbf{D}_t \\ \underline{\mathbf{D}}_{t+1} - \Lambda_t' \Pi_z' \underline{\mathbf{D}}_t \\ a_t - a_t^* \\ \forall t \in \{0, 1, \dots\}, \text{ given } \underline{\mathbf{D}}_0 \end{bmatrix} = \mathbf{0}$$

- **Note:** Much larger system compared to Ramsey due to last 2 eqs.

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- **Note:** Much larger system compared to Ramsey due to last 2 eqs.
 - $\mathbf{D}_t, a_t^*$ define mass and optimal savings policy at **the individual level**
 - Standard Ramsey model: 8 eqs. per period
 - HANC with $N_z = 7, N_a = 500$: $8 + 2 \times 7 \times 500 = 7008$ per period

Market clearing

- **Capital market:** $K_t = A_t = \int a_t^*(z_{it}, a_{it-1}) d\mathbf{D}_t$
- **Labor market:** $L_t = \int \ell_t^*(z_{it}, a_{it-1}) d\mathbf{D}_t = \int z_{it} d\mathbf{D}_t = 1$
- **Goods market:** $Y_t = \int c_t^*(z_{it}, a_{it-1}) d\mathbf{D}_t + I_t$
- **Walras:** Capital and labor market clears $\Rightarrow$ goods market clears (using Euler's theorem)

$$\begin{aligned} C_t^{hh} + I_t &= \int c_{it}^* d\mathbf{D}_t + [K_t - (1 - \delta)K_{t-1}] \\ &= \int [(1 + r_t)a_{it-1} + w_t z_{it} - a_{it}] d\mathbf{D}_t \\ &= [(1 + r_t)K_{t-1} + w_t L_t - K_t] + [K_t - (1 - \delta)K_{t-1}] \\ &= r_t^K K_{t-1} + w_t L_t \\ &= Y_t \end{aligned}$$

Computing the Stationary Equilibrium

Stationary equilibrium - equation system

The **stationary equilibrium** satisfies

$$\begin{bmatrix} r_{ss}^K - F_K(\Gamma_{ss}, K_{ss}, L_{ss}) \\ w_{ss} - F_L(\Gamma_{ss}, K_{ss}, L_{ss}) \\ r_{ss} - (r_{ss}^K - \delta) \\ A_{ss} - K_{ss} \\ A_{ss} - A_{ss}^{hh} \\ L_{ss} - L_{ss}^{hh} \\ A_{ss}^{hh} - \int a_{ss} d\mathbf{D}_{ss} \\ L_{ss}^{hh} - \int \ell_{ss} d\mathbf{D}_{ss} \\ \underline{\mathbf{D}}_{ss} - \Lambda'_{ss} \Pi'_Z \underline{\mathbf{D}}_{ss} \\ a_{ss} - a_{ss}^* \end{bmatrix} = \mathbf{0}$$

Note : Households still move around »inside« the distribution due to idiosyncratic shocks. Does not affect aggregates due to »law of large numbers«

Stationary equilibrium - more verbal definition

Given technology Γ_{ss}

1. Quantities K_{ss} and L_{ss} ,
2. prices r_{ss} and w_{ss} (always $\Pi_{ss} = 0$),
3. the distribution $\mathbf{D}_{ss}$ over z_{it} and a_{it-1}
4. and the policy functions a_{ss}^* , ℓ_{ss}^* and c_{ss}^*

are such that

1. Households maximize expected utility (policy functions)
2. Firms maximize profits (prices)
3. $\mathbf{D}_{ss}$ is the invariant distribution implied by the household problem
4. Mutual fund balance sheet is satisfied
5. The capital market clears
6. The labor market clears
7. The goods market clears

How do we solve the household block in practice?

The hard part is to solve the household block! How do we do it?

1. Use EGM to obtain the policy functions a_{ss}, c_{ss} for a given r, w
 - The »backward« step

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1. Use EGM to obtain the policy functions a_{ss}, c_{ss} for a given r, w
 - The »backward« step
2. Use the histogram method to obtain the stationary distribution $\mathbf{D}_{ss}$
 - The »forward« step
3. Aggregate policy:

$$A_{ss}^{hh}(\{r_{ss}, w_{ss}\}) = \int a_{ss}^*(z_{it}, a_{it-1}, \{r_{ss}, w_{ss}\}) d\mathbf{D}_{ss}$$

Direct implementation (K guess)

Technology: $F(K, L) = \Gamma K^\alpha L^{1-\alpha}$

Root-finding problem in K_{ss} with the objective function:

1. Set $L_{ss} = 1$ (and $\Pi_{ss} = 0$)
2. Calculate $r_{ss} = \alpha \Gamma_{ss} (K_{ss})^{\alpha-1} - \delta$ and $w_{ss} = (1 - \alpha) \Gamma_{ss} (K_{ss})^\alpha$
3. Solve infinite horizon household problem *backwards*, i.e. find $\mathbf{a}_{ss}^*$
4. Simulate households *forwards* until convergence, i.e. find $\mathbf{D}_{ss}$
5. Return $K_{ss} - \mathbf{a}_{ss}^{*'} \mathbf{D}_{ss}$

Note: $\mathbf{a}_{ss}^{*'} \mathbf{D}_{ss} = \sum_i a_{i,ss}^* D_i$

Direct implementation (r guess)

Technology: $F(K, L) = \Gamma K^\alpha L^{1-\alpha}$

Root-finding problem in r_{ss} with the objective function:

1. Set $L_{ss} = 1$ (and $\Pi_{ss} = 0$)
2. Calculate $K_{ss} = \left(\frac{r_{ss} + \delta}{\alpha \Gamma_{ss}} \right)^{\frac{1}{\alpha-1}}$ and $w_{ss} = (1 - \alpha) \Gamma_{ss} (K_{ss})^\alpha$
3. Solve infinite horizon household problem *backwards*, i.e. find $\mathbf{a}_{ss}^*$
4. Simulate households *forwards* until convergence, i.e. find $\mathbf{D}_{ss}$
5. Return $K_{ss} - \mathbf{a}_{ss}^{*'} \mathbf{D}_{ss}$

How to find a root numerically?

- Finding a root of a function means finding x^* such that $f(x^*) = 0$
- How do we do it on a computer? Two main solutions (in the univariate case)
 1. Bisection
 2. Newton's method

Root-finding: bisection

Equilibrium = root-finding problem: find x^* such that $f(x^*) = 0$

- Here: $f(K) = A_{ss}^{hh}(r(K), w(K)) - K$
- Each evaluation of f is costly: a full solve of the household block

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Bisection: start from a bracket $[\underline{x}, \bar{x}]$ with $f(\underline{x})f(\bar{x}) < 0$

1. Compute the midpoint $m = (\underline{x} + \bar{x})/2$
2. Stop if $|f(m)| < \text{tol}$
3. Keep the half where the sign changes: if $f(\underline{x})f(m) < 0$ set $\bar{x} = m$, else set $\underline{x} = m$

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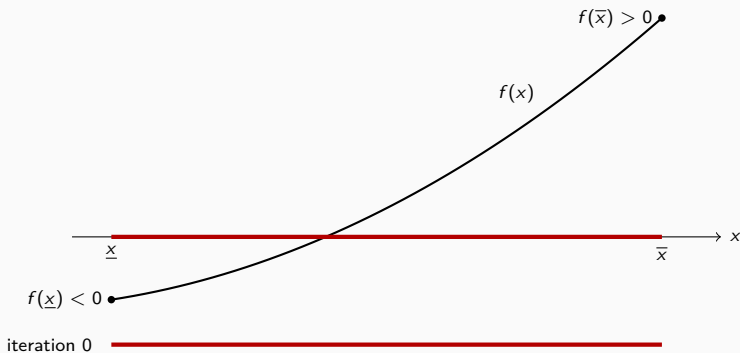
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Properties:

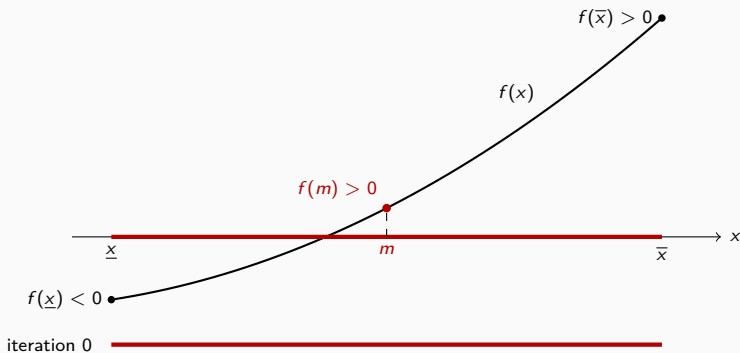
- Always converges if f is continuous
- Slow: the bracket halves each iteration, error $\leq (\bar{x} - \underline{x})/2^n$
- Only works in one dimension

Bisection: intuition



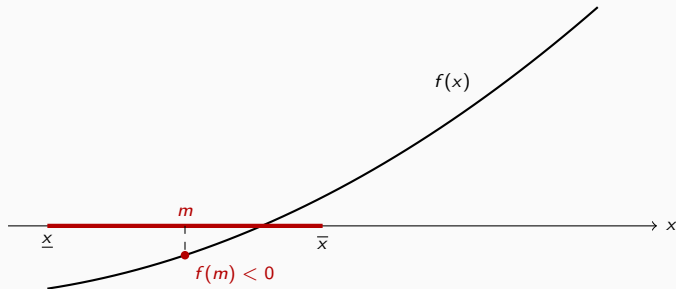
f changes sign on $[\underline{x}, \bar{x}]$, so if f is continuous there is a root inside.

Bisection: intuition



$f(m) > 0$, same sign as $f(\bar{x})$: the root is to the left of m . Set $\bar{x} = m$.

Bisection: intuition

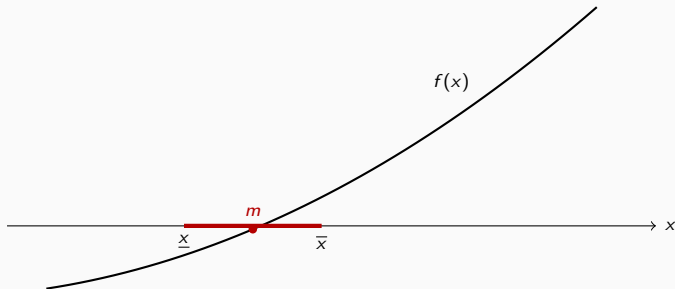


iteration 0

iteration 1

$f(m) < 0$, same sign as $f(\underline{x})$: the root is to the right of m . Set $\underline{x} = m$.

Bisection: intuition



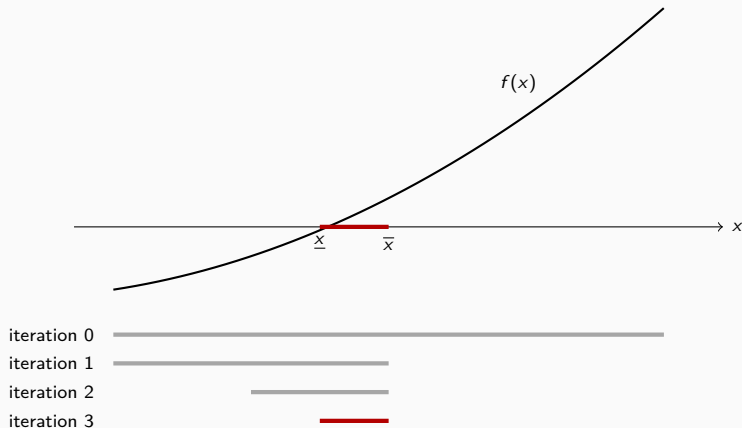
iteration 0 

iteration 1 

iteration 2 

Again $f(m) < 0$: set $\underline{x} = m$.

Bisection: intuition



Each evaluation of f cuts the bracket in half, whatever the shape of f .

Root-finding: Newton

Newton: linearize f around the guess x_n and solve the linear equation

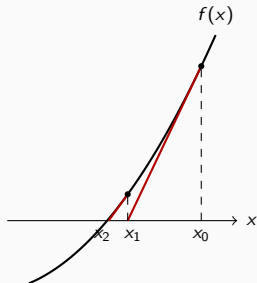
$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

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No analytical derivative? One extra evaluation of f : $f'(x) \approx \frac{f(x+h)-f(x)}{h}$

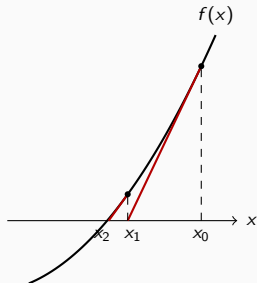


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Properties:

- Fast: very few iterations close to the root
- Not safe: can diverge or leave the domain from a bad guess
- Works in many dimensions (transition path, next lectures):

$$\mathbf{x}_{n+1} = \mathbf{x}_n - \mathbf{J}(\mathbf{x}_n)^{-1} \mathbf{f}(\mathbf{x}_n)$$

Let's code!

Technology: $F(K, L) = \Gamma K^\alpha L^{1-\alpha}$

Consider Γ_{ss} and δ as »free« parameters:

1. Choose r_{ss} and w_{ss}
2. Solve infinite horizon household problem *backwards*, i.e. find $\mathbf{a}_{ss}^*$
3. Simulate households *forwards* until convergence, i.e. find $\mathbf{D}_{ss}$
4. Set $K_{ss} = \mathbf{a}_{ss}^{*'} \mathbf{D}_{ss}$
5. Set $L_{ss} = 1$ (and $\Pi_{ss} = 0$)
6. Set $\Gamma_{ss} = \frac{w_{ss}}{(1-\alpha)(K_{ss})^\alpha}$
7. Set $r_{ss}^K = \alpha \Gamma_{ss} (K_{ss})^{\alpha-1}$
8. Set $\delta = r_{ss}^k - r_{ss}$

Direct implementation (β)

Set $r_{ss} = r^{target}$, $K_{ss} = K^{target}$, $Y_{ss} = Y^{target}$, $w_{ss}L_{ss}/Y_{ss} = s_L^{target}$ and back out

1. $\alpha = 1 - s_L^{target}$
2. $\Gamma_{ss} = Y^{target} / (L_{ss}^{1-\alpha} K_{ss}^{\alpha})$
3. $\delta = \alpha Y^{target} / K^{target} - r^{target}$

We know that $w_{ss} = (1 - \alpha)Y^{target}$. Then find the β that clears the market

Root-finding problem in β with the objective function:

1. Set $L_{ss} = 1$ (and $\Pi_{ss} = 0$),
2. Solve infinite horizon household problem *backwards*, i.e. find $\mathbf{a}_{ss}^*$ for a given β
3. Simulate households *forwards* until convergence, i.e. find $\mathbf{D}_{ss}$
4. Return $K_{ss} - \mathbf{a}_{ss}^{*'} \mathbf{D}_{ss}$

How to choose parameters?

- **External calibration:** Set subset of parameters to the *standard values in the literature* or *directly from data estimates* (e.g. income process)

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 1. **Informal:** Roughly match targets by hand
 2. **Formal:**
 - 2a. Solve root-finding problem
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- **Complication:** *We must always solve for the steady state for each guess of the parameters to be calibrated*

Calibration at the quarterly level

- $r_{ss} = 0.05/4$ to match 5% annual interest rate
- $Y_{ss} = 1$ as a normalization
- $K_{ss} = 16$ to match annual wealth-to-output ratio of 4
- $\alpha = 1/3$ to match labor share of roughly $2/3$
- $\sigma_\psi = 0.5$, $\rho = 0.9$: data on income inequality and risk

Some Properties of the HANC steady-state

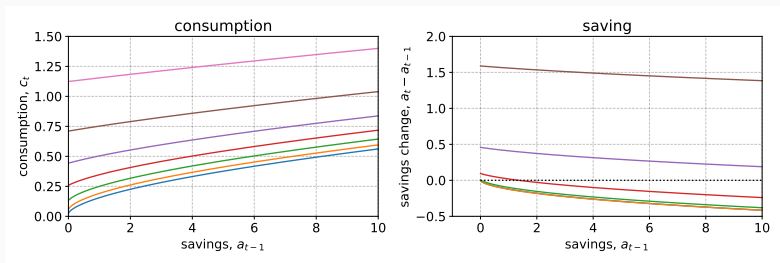
Consumption function

- Euler-equation still necessary for $a_{it} > 0$:

$$c_{it}^{-\sigma} = \beta(1 + r_{t+1})\mathbb{E}_t [c_{it+1}^{-\sigma}]$$

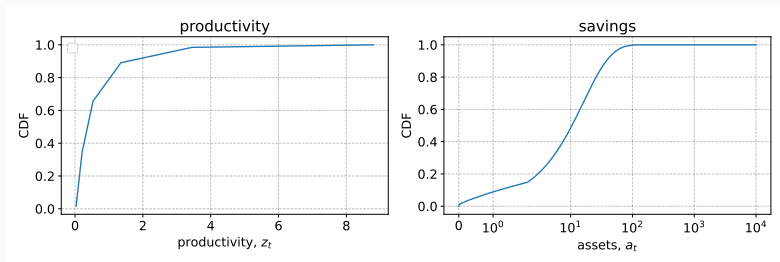
- Precautionary saving:

1. Low consumption for low cash-on-hand $\rightarrow$ *buffer-stock target*
2. Steep slope for low cash-on-hand $\rightarrow$ *high MPC*



Some amount of inequality

- **Productivity:** Marginal distribution over only z_{it}
- **Savings:** Marginal distribution over only a_{it}



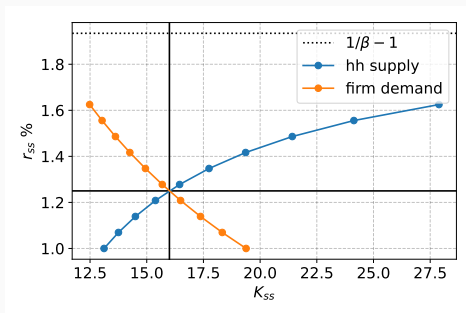
- **Drivers of wealth inequality here: income shocks**

Steady state interest rate

- **Representative agent / complete markets:**

Derived from aggregate Euler-equation

$$C_t^{-\sigma} = \beta(1 + r_{t+1})C_{t+1}^{-\sigma} \Rightarrow C_{ss}^{-\sigma} = \beta(1 + r_{ss})C_{ss}^{-\sigma} \Leftrightarrow \beta = \frac{1}{1 + r_{ss}}$$



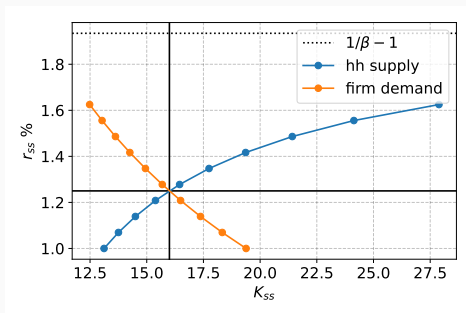
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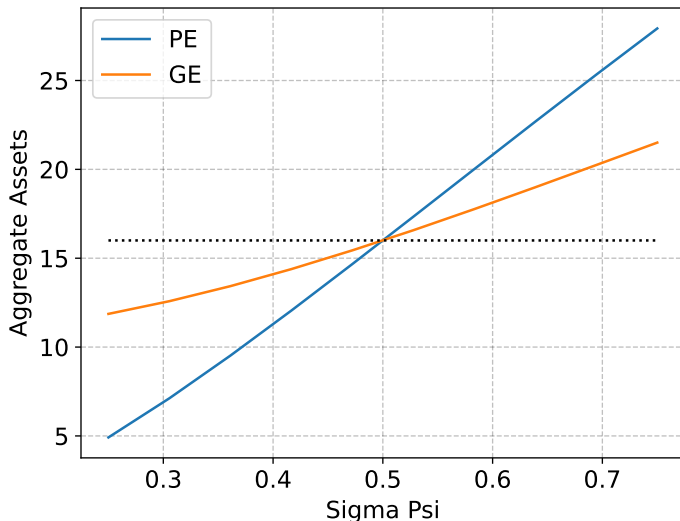
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- **Heterogeneous agents:** *No such equation exists*

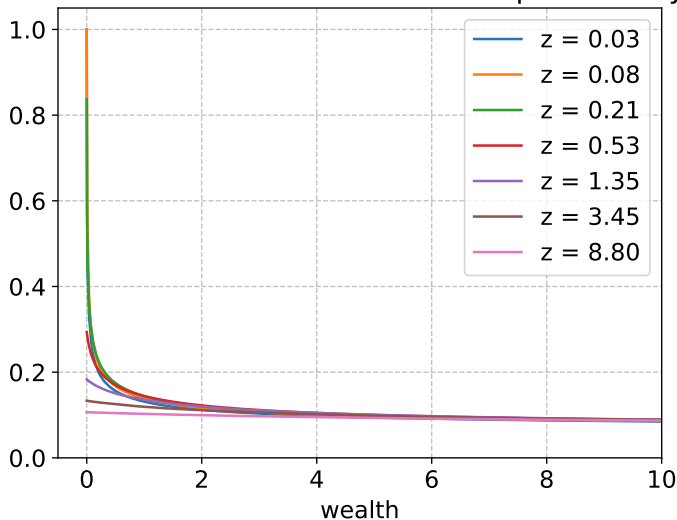


Risk drives wealth accumulation

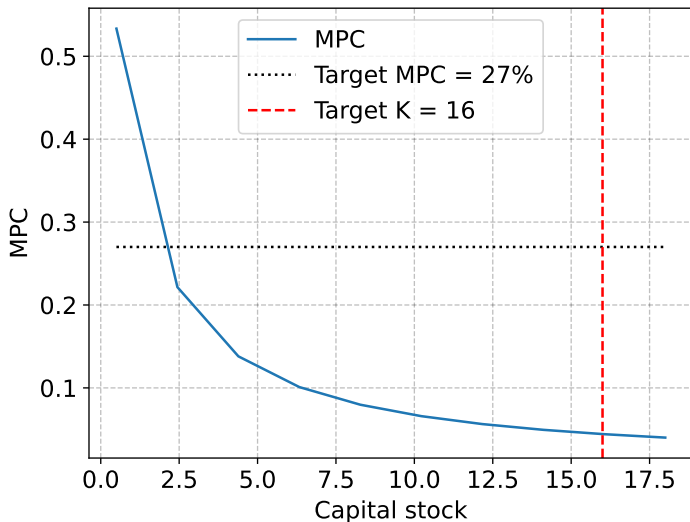


Marginal Propensity to Consume

MPCs as a function of wealth and productivity



Tradeoff between matching aggregate wealth and MPCs



Exercises

Exercise 1: HANC with ex-ante heterogeneity

Add permanent β heterogeneity to the HANC model:

$$\begin{aligned} v_t(\beta_i, z_{it}, a_{it-1}) &= \max_{c_{it}} \frac{c_{it}^{1-\sigma}}{1-\sigma} + \beta_i \mathbb{E}_t [v_{it+1}(z_{it+1}, a_{it})] \\ \text{s.t. } a_{it} + c_{it} &= (1 + r_t)a_{it-1} + w_t z_{it}, \quad a_{it} \geq 0 \\ \log z_{it+1} &= \rho_z \log z_{it} + \psi_{it+1}, \quad \psi_{it} \sim \mathcal{N}(\mu_\psi, \sigma_\psi), \quad \mathbb{E}[z_{it}] = 1 \end{aligned}$$

Assume that we have three types of households:

$\beta_i \in \{\beta - \delta^\beta, \beta, \beta + \delta^\beta\}$. Find δ^β such that $MPC = 0.27$ and $K/Y = 16$.

Exercise 2: HANCGovModel

- **No production.** No physical savings instrument
- **Households:** Get stochastic endowment z_{it} of consumption good
- **Government:**
 1. Choose government spending
 2. Collect taxes, τ_t , proportional to endowment
 3. Bonds: Pays 1 unit of the consumption good next period. Price is $p_t^B < 1$

$$p_t^B B_t + \int \tau_t z_{it} d\mathbf{D}_t = B_{t-1} + G_t$$
$$\tau_t = \tau_{ss} + \varphi(B_{t-1} - B_{ss})$$

- **Market clearing:**

$$B_t = A_t^{hh}$$
$$C_t^{hh} + G_t = \int z_{it} d\mathbf{D}_t = 1$$

Exercise 2: Households

Households:

$$v_t(z_{it}, a_{it-1}) = \max_{c_{it}} \frac{c_{it}^{1-\sigma}}{1-\sigma} + \beta \mathbb{E}_t [v_{it+1}(z_{it+1}, a_{it})]$$

$$\text{s.t. } p_t^B a_{it} + c_{it} = a_{it-1} + (1 - \tau_t) z_{it}, \quad a_{it} \geq 0$$

$$\log z_{it+1} = \rho_z \log z_{it} + \psi_{it+1}, \quad \psi_{it} \sim \mathcal{N}(\mu_\psi, \sigma_\psi), \quad \mathbb{E}[z_{it}] = 1$$

Euler-equation:

$$c_{it}^{-\sigma} = \beta \frac{v_{a,t+1}(z_{it}, a_{it})}{p_t^B}$$

Envelope condition:

$$v_{a,t}(z_{it-1}, a_{it-1}) = c_{it}^{-\sigma}$$

Exercise 2: Questions

1. **Define the stationary equilibrium**
2. **Solve and simulate the household problem**
with $p_{ss}^B = 0.975$ and $\tau_{ss} = 0.12$.
3. **Find the stationary equilibrium**
with $G_{ss} = 0.10$ and $\tau_{ss} = 0.12$.
4. **What happens for $\tau_{ss} \in (0.11, 0.15)$?**
5. **When is average household utility maximized?**

Note: Full solution available at this [link](#)